

BIOINFORMATICS

WEEK 2 – DAY 14

Protein-Protein Interaction (PPI) is the specific physical contact that occurs when two or more protein molecules bind together. Proteins rarely work alone; instead, they attach to each other to form physical complexes that drive almost every process inside a living cell.

Key Aspects of PPIs

- **Types:** Interactions can be permanent (forming long-lasting molecular machines like ribosomes) or transient (briefly touching to pass a chemical signal).
- **Composition:** They can occur between identical proteins (homotypic) or completely different proteins (heterotypic).
- **Function:** PPIs control cell signaling, regulate metabolic enzymes, and build the physical shape and scaffolding of cells.
- **The Interactome:** The complete global map of all protein interactions within an organism is called its interactome.

WHY STUDY PROTEIN NETWORKS?

1. Sees the Whole Picture

- **Shifts focus:** Moves from single proteins to entire cell systems.
- **Maps connections:** Charts the complete global interactome web.

2. Predicts Unknown Functions

- **Guilt by association:** Infers unknown protein roles via connected neighbors.
- **Finds core modules:** Groups interacting proteins into functional cellular machines.

3. Explains Disease Mechanisms

- **Spots network breakdowns:** Links genetic mutations to collapsed biological pathways.
- **Contextualizes oncogenes:** Explains how alterations in TP53 or EGFR drive cancer.

4. Directs Drug Discovery

- **Identifies targets:** Pinpoints crucial network hubs to disable diseases.
- **Foresees side effects:** Predicts how blocking one node impacts neighboring proteins.

5. Optimizes Computational Data

- **Integrates omics:** Combines experimental big data using the STRING database.
- **Models networks:** Simulates cellular disruption inside Cytoscape before wet-lab testing.

STRING stands for Search Tool for the Retrieval of Interacting Genes/Proteins.

Core Facts About STRING

- **Purpose:** It is a massive biological database and web resource that catalogs known and predicted protein-protein interactions.
- **Scope:** It includes both direct physical interactions and indirect functional associations.
- **Data Sourcing:** It aggregates data from primary experimental repositories, computational prediction methods (like genomic context), public text-mining, and co-expression data.
- **Output:** It scores the confidence of interactions and generates network graphs that can be exported directly into software like Cytoscape.

CONFIDENCE SCORES

Interactions are usually shown with scores between **0** and **1**.

In the [STRING database](#), a Confidence Score is a numerical value that indicates the probability or likelihood that a predicted protein-protein interaction is true.

These scores do not measure physical binding strength or the biological specificity of the interaction. Instead, they quantify the database's certainty to help researchers filter out biological noise and false positives.

- **Low Confidence (≥ 0.150):** Highly exploratory, includes weak evidence channels like computational predictions or loose text-mining.
- **Medium Confidence (≥ 0.400):** The **default setting** on STRING; balances biological sensitivity with specificity.
- **High Confidence (≥ 0.700):** Conservative threshold; standard for rigorous pathway or protein mapping.
- **Highest Confidence (≥ 0.900):** Extremely stringent; mostly retains interactions backed by solid experimental evidence or curated databases.

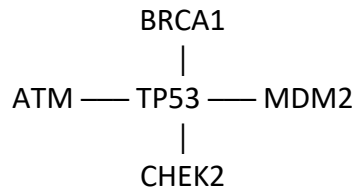
Individual Evidence Channels

A combined confidence score is calculated by probabilistically merging data from independent sources (channels). Each channel carries its own sub-score:

- **Experimental (escore):** Direct evidence from laboratory assays like yeast two-hybrid or mass spectrometry.
- **Database (dscore):** Manually curated pathways from source repositories like KEGG or Reactome.
- **Text-mining (tscore):** Natural Language Processing (NLP) extracting co-occurrences of protein names in literature.
- **Coexpression (ascore):** Correlated RNA or protein expression levels across datasets.

- **Genomic Context:** Composed of **Neighborhood** (conserved gene proximity), **Fusion** (merged genes), and **Co-occurrence** (presence/absence across species).

UNDERSTANDING A NETWORK



- TP53 is connected to several proteins. This suggests it plays a central role.
- Each circle is called a node.
- Each connecting line is called an edge.

HUB PROTEIN is a highly connected protein positioned at the center of a biological network.

Instead of interacting with just one or two partners, a hub protein binds to dozens or hundreds of different proteins, serving as a critical communication traffic controller inside the cell

They are often:

- Essential for survival.
- Involved in major cellular pathways.
- Important drug targets.

TP53 is an example of a hub protein.

Key Characteristics of Hubs

- **High Degree Connectivity:** They possess the highest number of edges (connecting lines) in an interactome map [20.40].
- **Biological Essentiality:** They are usually indispensable for cell survival. Deleting a hub gene is often lethal to an organism.
- **Centrality:** They act as bridges that link distinct cellular pathways and functional modules together.
- **The TP53 Example:** In your network, TP53 acts as a classic hub protein because it centrally connects to ATM, MDM2, CHEK2, and BRCA1 to control cell division and DNA repair.